

32-bit microcontroller

TIM3 wave-by-wave current limiting function

Application Note

Rev1.01 June 2025

Applicable objects

Product series	Product model	Product series	Product model
L series	HC32L130 HC32L136 HC32L072 HC32L073 HC32L170 HC32L176 HC32L190 HC32L196 HC32L166 HC32L186	F series	HC32F030 HC32F072 HC32F170 HC32F176 HC32F190 HC32F196 HC32F002 HC32F052 HC32F420
A series	HC32A052	-	-

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1 Overview

This document introduces the wave-by-wave current limiting (voltage limiting) function of the timer TIM3 (ATIM3) of a specific series of MCUs from Xiaohua Semiconductor.

2 Wave-by-wave current limiting function of **TIM3** timer

2.1 Protection measures of **TIM3** timer **PWM**

TIM3 can output independent or complementary PWM. When the output PWM is used for power applications such as modulating current (voltage) or controlling motor drive, the control circuit will have a need for protection measures to shut down PWM in an emergency (making the timer PWM pin output a specified level, change to high impedance, etc.).

Depending on the degree of shutting down PWM in an emergency, TIM3 has two protection measures: braking and wave-by-wave current limiting (voltage limiting). This article mainly introduces the wave-by-wave current limiting (voltage limiting) function.

2.2 Wave-by-wave current limiting function

The so-called wave-by-wave current limiting refers to the real-time monitoring of the inductor current or the power tube bridge drive circuit during the operation of the power supply or motor drive circuit. Once the current exceeds the set threshold, the PWM signal is immediately turned off in the current PWM cycle until the next event updates UEV and then automatically enables the PWM output. If the current signal continues to exceed the threshold, the PWM signal is turned off until the monitored current returns to below the threshold. Then, after the next event updates UEV, the PWM resumes output. As shown in Figure 2-1:

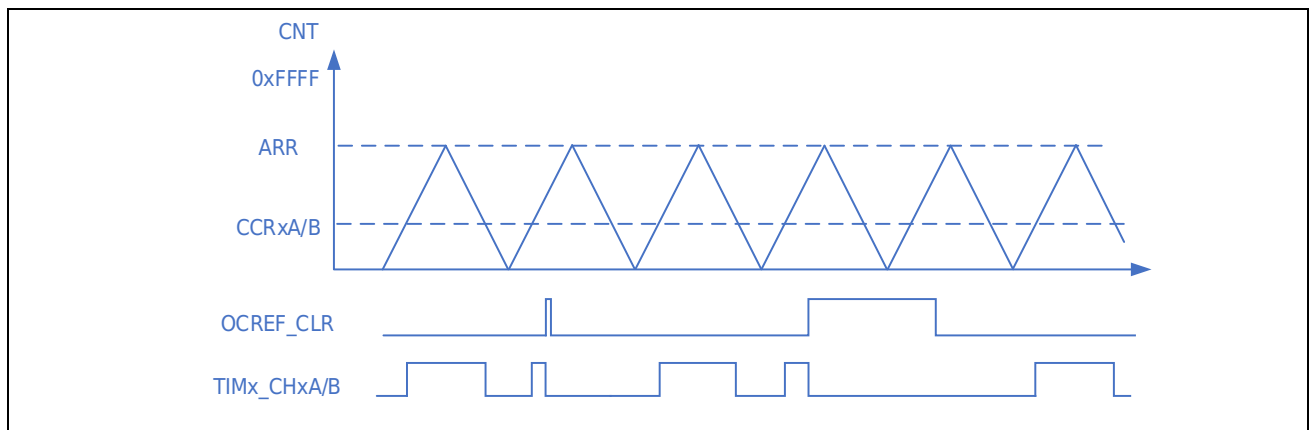


Figure 2-1 Principle of wave-by-wave current limiting function

2.3 Configuration of wave-by-wave current limiting function

TIM3 can receive monitoring circuit signals:

- REFCLR output by VC comparator
- ETFP signal after filtering and phase selection at ETP pin

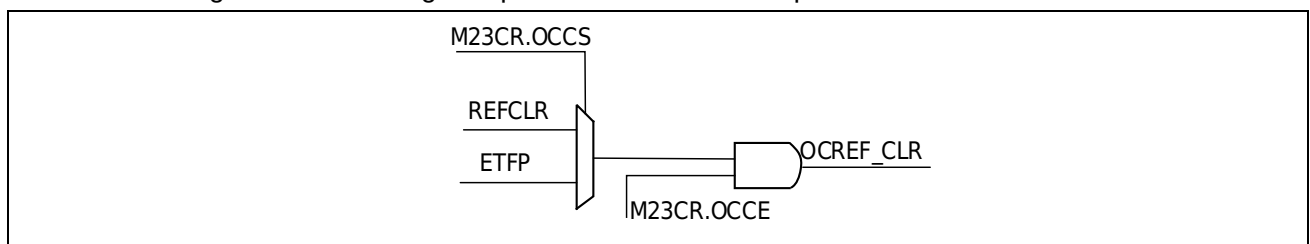


Figure 2-2 Wave-by-wave current limiting signal

When the current exceeds the set threshold, the monitoring circuit outputs a signal to the receiving port of the wave-by-wave current limiting function of TIM3, and TIM3 will give an OCREF_CLR signal. When the wave-by-wave current limiting function is enabled, the PWM output of the current cycle is turned off until the next event updates UEV and the PWM starts to resume output.

VC's OUT_CFG.TIM3RCLR sets whether to enable the REFCLR signal to TIM3.

OUT_CFG.INV_TIMER determines whether the REFCLR signal is high level active (OUT_CFG.INV_TIMER=0) or low level active (OUT_CFG.INV_TIMER=1).

FLTR.ETP sets whether the input level of the ETR port is high level active (FLTR.ETP=0) or low level active (FLTR.ETP=1), and FLTR.FLTET sets the filter time of ETR.

When DTR.AOE=0, DTR.MOE=1, after CR.OCCE is enabled, the valid OCREF_CLR signal can clear the current OCREFxy ($x=0/1/2$, $y=A/B$) signal (this clearing action is valid when $FLTR.OCMyx > 1$ ($x=0/1/2$, $y=A/B$)). After OCREFxy is cleared, the corresponding PWM output is turned off, and DTR.MOE will not be cleared. When the OCREF_CLR signal is invalid, the PWM continues to output after the next event updates UEV.

An event update UEV cycle can be one TIM3 timer cycle, or several TIM3 timer cycles, depending on the configuration of the valid cycle register.

2.4 Wave-by-wave current limiting with interval period of 0

When TIM3 outputs independent or complementary PWM, the wave-by-wave current limiting function can be used.

In the following figure, the yellow channel 1 and the blue channel 2 are TIM3_CH0A and TIM3_CH0B respectively, and the red channel 3 is the input signal on the TIM3_ETR pin. In the program, OCREF_CLR is set from ETRP and is high effective, and TIM3 is in triangle wave mode.

When TIM3 outputs independent PWM, when a high-level pulse appears on the ETR pin, TIM3_CH0A and TIM3_CH0B are both turned off in the current PWM cycle, and then automatically resume output in the next PWM cycle (the RCR register configuration interval period is 0). As shown in the figure:

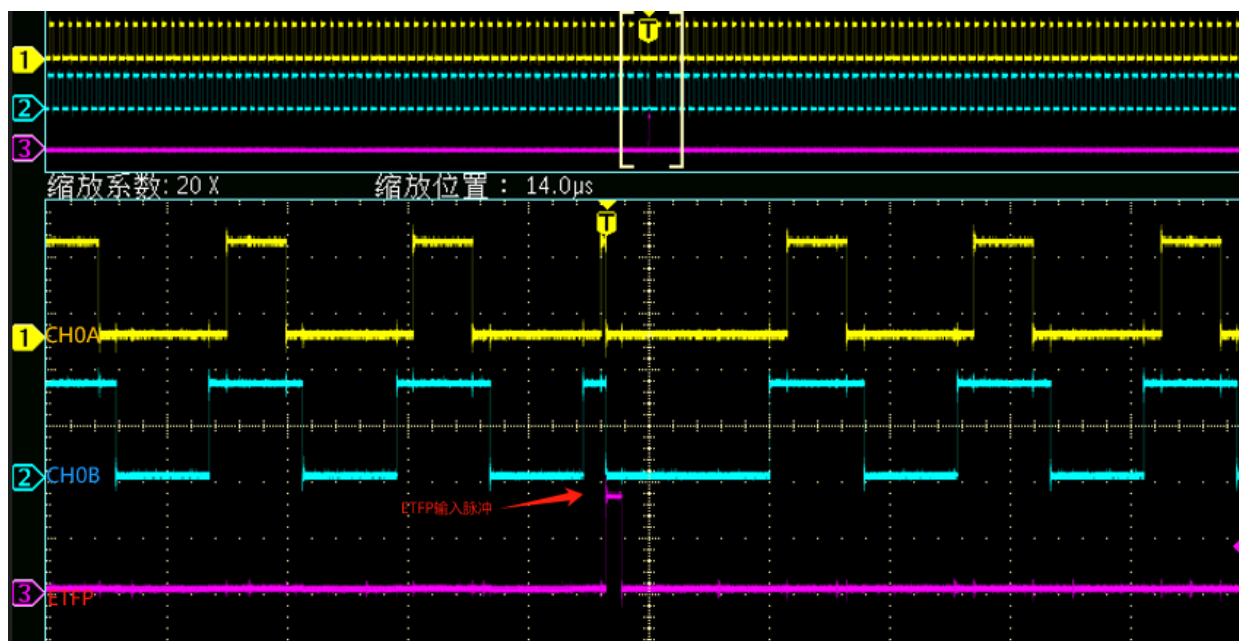


Figure 2-3 Independent PWM wave-by-wave current limiting

When TIM3 is configured to output complementary PWM with dead zone (high level is valid), when a high level pulse appears on the ETR pin, TIM3_CH0A outputs PWM and is turned off and outputs low, while TIM3_CH0B outputs PWM and outputs high level. It should be noted that when TIM3_CH0A is turned off due to the ETRP high level pulse, TIM3_CH0B does not immediately output a high level, but waits for the configured dead time to output a high level.

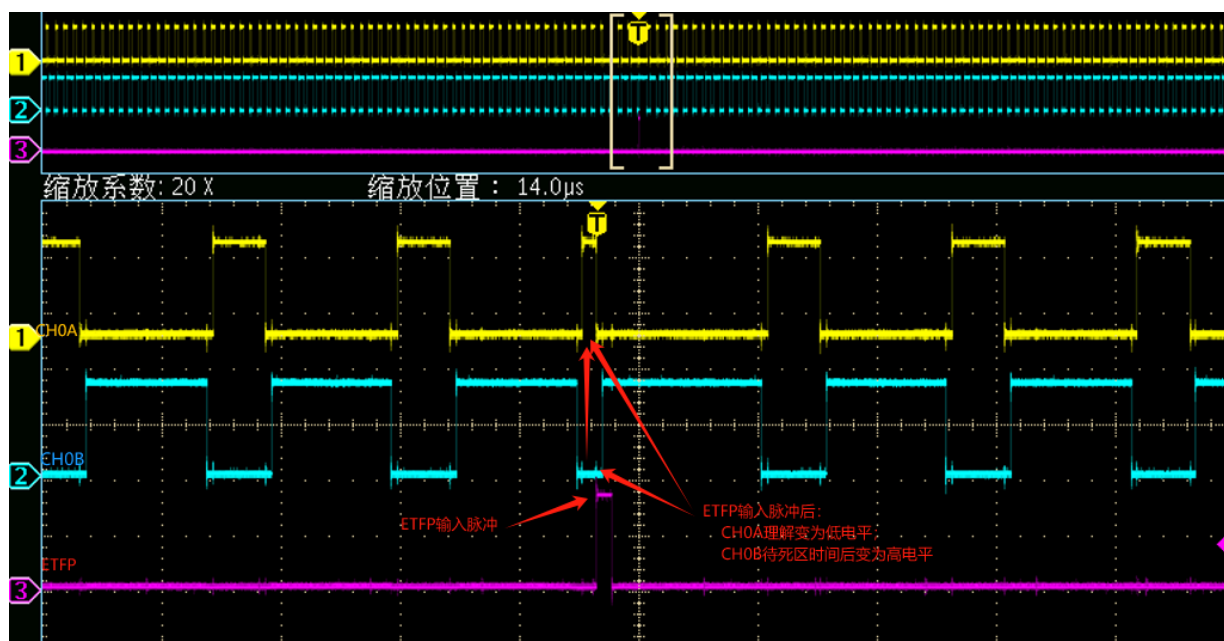


Figure 2-4 Dead-zone complementary PWM wave-by-wave current limiting

2.5 Wave-by-wave current limiting with interval period of 1

When the RCR register is configured with an interval period of 1, the event update UEV occurs every 2 timer cycles. Since the comparison values of each output are set with a cache function, the duty cycle change of the PWM takes effect after the event update UEV, so the modulation of the PWM duty cycle also changes to every two timer cycles.

As shown in Figure 2-5, the duty cycle of TIM3_CH0A is originally the same for both. When the ETFP pulse occurs in the first cycle, the output of TIM3_CH0A is turned off, and then TIM3_CH0B outputs a high level. The shutdown time of TIM3_CH0A lasts until the end of the second cycle, until the next event update UEV occurs, and the PWM resumes output.

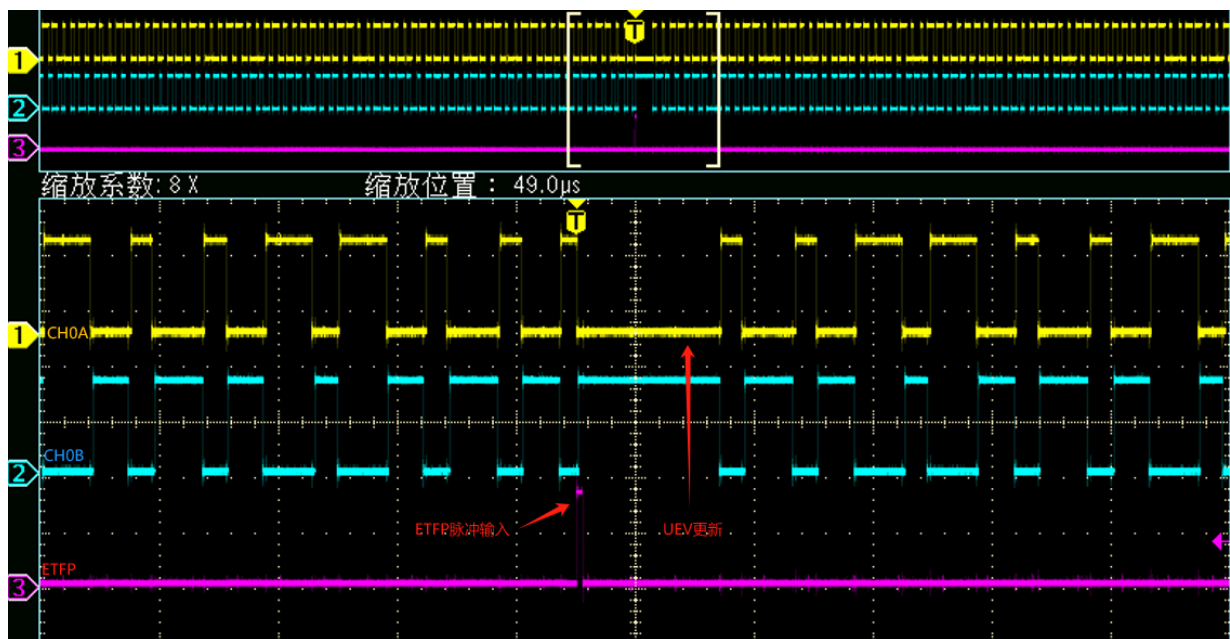


Figure 2-5 Wave-by-wave current limiting with an interval period of 1 (1)

If the ETFP pulse occurs during the second timer cycle, the output of TIM3_CH0A is also turned off, and then TIM3_CH0B outputs a high level. The off time of TIM3_CH0A lasts until the next event updates UEV, because the off time shown in Figure 2-6 is obviously shorter than that in Figure 2-5.

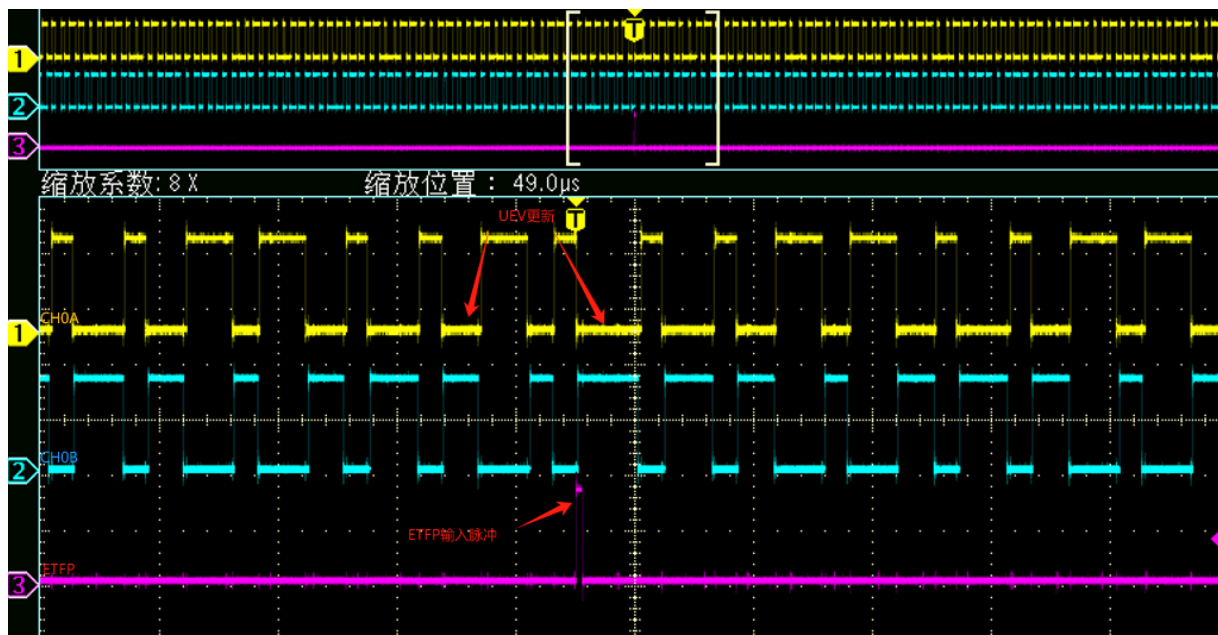


Figure 2-6 Wave-by-wave current limiting with an interval period of 1 (2)

If the ETFP pulse occurs during the TIM3_CH0A off period at the end of the second timer cycle, if the ETFP pulse level time is shorter than the off time of TIM3_CH0A, then the PWM resumes output after the next event updates UEV. From the waveform, it seems that TIM3_CH0A is not turned off. As shown in Figure 2-7:

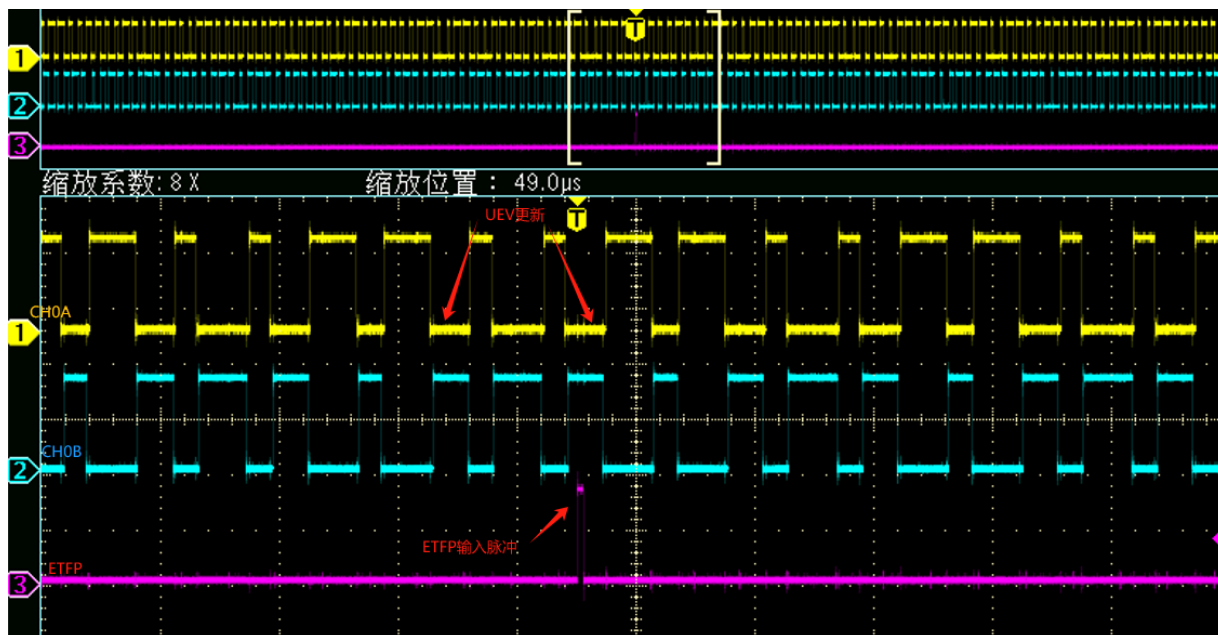


Figure 2-7 Wave-by-wave current limiting with an interval period of 1 (3)

If the signal width of OCREF_CLR spans multiple cycles, the PWM signal of CHxA will be turned off for the same number of cycles until the next event updates UEV and the output is restored.

2.6 Wave-by-wave current limiting function of **TIM012**

TIM0/1/2 is the same as TIM3 and also has the wave-by-wave current limiting function. The implementation method is described above.

3 Summary

The above chapters briefly introduce the wave-by-wave current limiting function of Xiaohua Semiconductor MCU*. In the actual application development process, if the user needs to further understand the usage and operation of this module, the corresponding reference manual should prevail. The samples and driver libraries mentioned in this chapter can be used as further experiments and learning for users, and can also be used as references in actual development.

4 Reference examples and drivers

For the latest drivers and samples of the relevant TIM3 modules, please refer to the corresponding MCU model driver library samples on the official website.

Revision history

Version	Date	Revision Content
Rev1.00	2025/01/20	Initial version released.
Rev1.01	2025/06/04	Applicable objects: HC32A052 series models added.